

Homework 2

Conceptual Design of a Quadrotor

Due Date: Feb 20th, 2026

The goal is to carry out conceptual design of a quadrotor that can carry out the following mission:

- Payload to be lifted is 300 grams
- Hover endurance desired is 20 minutes at sea level conditions (density of air = 1.225 kg/m^3)

You can make the following assumptions:

- The rotor blades have fixed pitch (blade root pitch can be used as input parameter to find out C_T). Assume constant chord.
- The linear twist distribution as function of non-dimensional radius (r) for blades of propellers is as follows: $\theta(r) = \theta_0 - 17 r$ (in degrees, make sure you convert to radian for calculations)
- Use the BEMT code developed in earlier homework to do the performance prediction for the rotor blades for design calculations.
- Once C_T is known, tip speed or RPM can be found to match hover thrust requirement and then design thrust (twice of hovering thrust).
- You can use the same airfoil properties as given in earlier homework (with or without lookup table).
- Use motor efficiency $\eta_m = 0.80$
- Baseline disc loading typical of such small aerial vehicle is in range of 40 to 90 N/m^2

The problem is to be solved by writing a computer simulation in language (preferably Matlab).

The computer program should iterate on the total weight till it converges to a final weight (the criteria for convergence could be that the difference of previous weight and current / new weight should be in range of 5 gram). Use all the weight formulas from paper (shared with earlier lecture) and power calculations using BEMT code. The following inputs have to be considered:

1. Endurance
2. Rotor Radius
3. Rotor operating C_T (this has to be set by selecting a root blade pitch angle, you can take baseline of 22 degrees and consider variations)
4. Solidity
5. Aspect Ratio of Blades
6. No. of Blades
7. No. of Battery Cells
8. Payload Weight
9. Initial GTOW (Assumed to be 3X Payload to begin with)

The unknown inputs need to be “guessed and varied” based on existing similar quadrotor drones and a justification should be given. Each guess has to be “changed” to see the effect of these guesses on the final design.

Output:

1. Show in a table details of all up weight of quadrotor with breakdown for each component (battery, motor, ESC, electronics, payload, airframe etc.) for any one design.
2. Give all the geometric and physical specifications (rotor diameter, chord, twist overall size, number of blades in each rotor etc.) for the quadrotor that you have designed (such as motor kV, current rating, battery capacity)
3. Explore the effect of various guesses (parameters) on the design (keeping other guesses constant). Objective is to identify optimal system with minimum GTOW. Decide how best to represent these results through various graphs.
4. Final minimum GTOW achieved through sweeping various guess parameters such as disc loading, number of blades in a rotor, number of cells, aspect ratio etc. and final parameters, compare with the non-optimal design shown in 1.